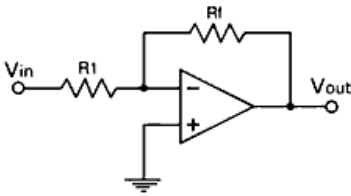


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer 2024 Course: B.Tech. Branch :Electronics and Computer Engineering Semester :III / Electronic and Computer Science Engineering Subject Code & Name: Electronics Devices & Circuits (BTECPC302) Max Marks: 60 Date:02/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the working of N - channel E - MOSFET with diagram and also draw the drain characteristics.	02/CO1	6
B)	Compare Common Source ,Common Drain and Common Gate configurations of FET .	02/CO1	6
C)	Discuss the different parameters of JFET.	02/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain the operation of Op -Amp Integrator with expression for its output voltage.	03/CO2	6
B)	Draw and explain the block diagram of Op - Amp in detail.	02/CO2	6
C)	Explain the inverting amplifier and Determine the voltage gain of inverting amplifier with $R_1 = 10K\Omega$ and $R_f = 47 K\Omega$.	03/CO2	6
			
Q. 3	Solve Any Two of the following.		12
A)	Explain the effect of negative feedback on any three characteristics of amplifier .	02/CO3	6
B)	Draw the circuit diagram of Wein bridge oscillator and calculate the frequency of oscillations with $R_1 = R_2 = 220 K\Omega$ and $C_1 = C_2 = 250 pF$.	03/CO3	6
C)	Explain the construction and working of RC phase shift oscillator .	02/CO3	6

Q.4	Solve Any Two of the following.		12
A)	Explain the operation of transistorized series voltage regulator .	02/CO4	6
B)	Describe the three terminal voltage regulators.	02/CO4	6
C)	Write a short note on voltage regulator IC 723.	02/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Discuss the Classification of Transducers .	02/CO5	6
B)	Explain construction and working of LVDT.	03/CO5	6
C)	Write a short note on Temperature Transducers.	02/CO5	6
	*** End ***		